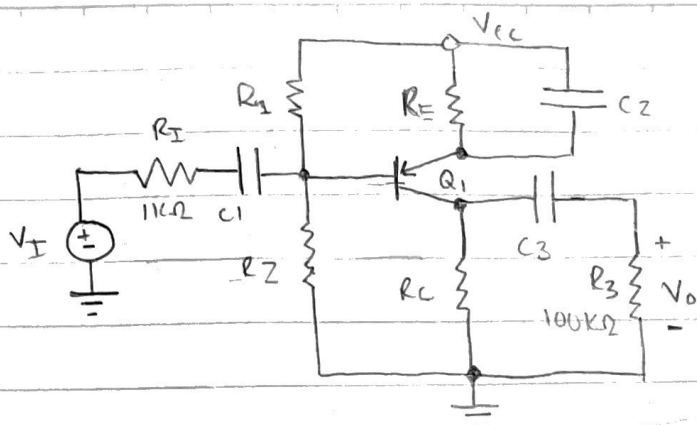


1. PNP Transistor → (DC Analysis)



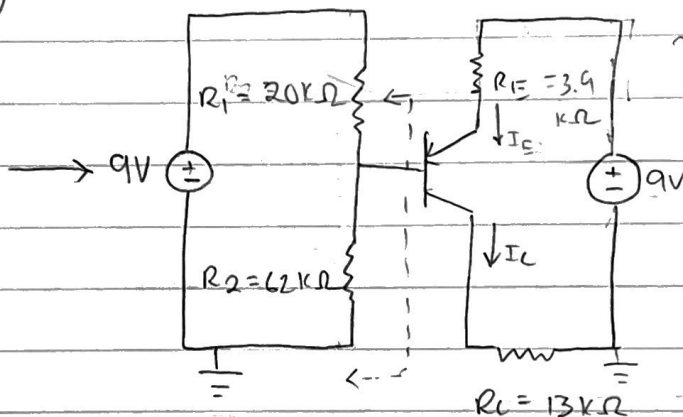
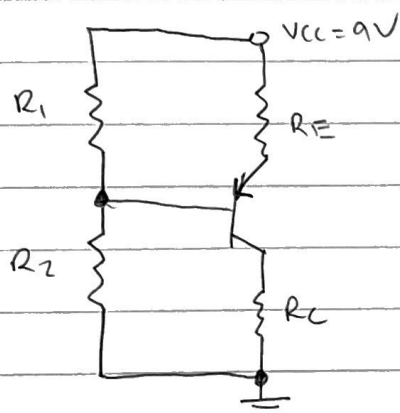
$$\beta_F = 135, V_{CC} = 9V$$

$$R_1 = 20k\Omega, R_2 = 62k\Omega$$

$$R_C = 13k\Omega, R_E = 3.9k\Omega$$

$$R_T = 1k\Omega, R_L = 100k\Omega$$

a) DC equivalent CKT → Capacitors are open



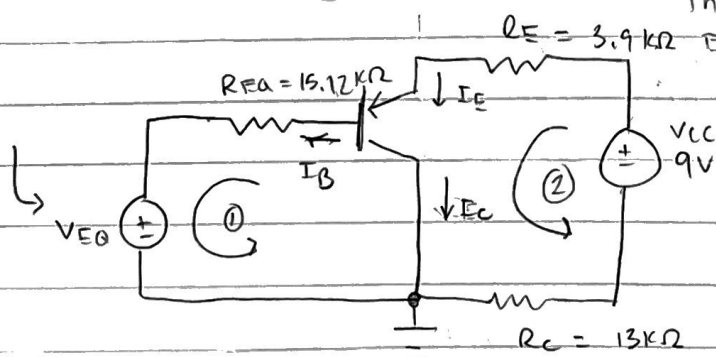
$$V_{EQ} = \frac{R_2}{R_1 + R_2} V_{CC}$$

$$\therefore V_{EQ} = 6.8V \sim$$

$$R_{EQ} = \frac{V_{EQ}}{V_{CC}/R_1} = \frac{R_1 R_2}{R_1 + R_2}$$

$$\therefore R_{EQ} = 15.12k\Omega \sim$$

Thevenin
Equivalent



$$\text{Loop 1: } (I_B R_{EQ}) + V_{EQ} - (I_C R_C) = 0$$

$$V_{EQ} = I_C R_C = I_B R_{EQ}$$

$$6.8V = 13k\Omega I_C = 15.12k\Omega I_B$$

$$I_C = \beta_F I_B = 135 I_B$$

$$6.8V = 1755 I_B + 15.12k\Omega I_B$$

$$\therefore I_B = 3.91 \mu A$$

$$I_C = 135 \times 3.91 \mu A = 0.528 mA$$

$$I_E = (135 + 1) I_B = 0.531 mA$$

$$\text{Loop 2: } -V_{CC} + (R_E I_E) + V_{EC} + (I_C R_C) = 0$$

$$V_{EC} = V_{CC} - R_E I_E - I_C R_C$$

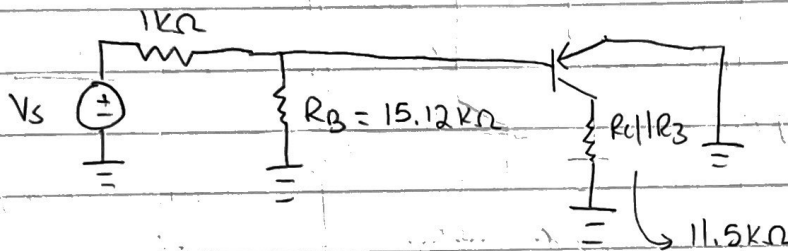
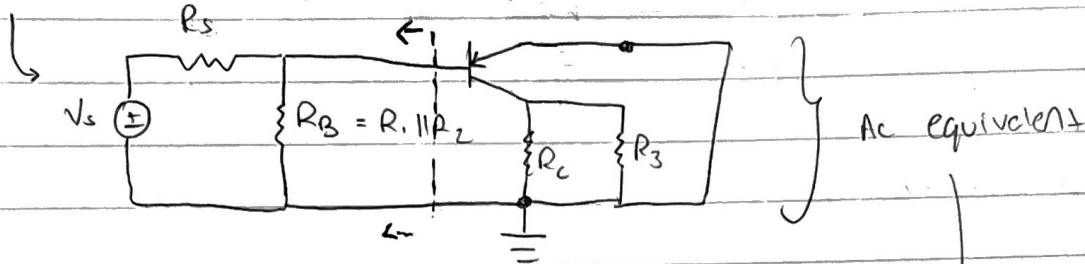
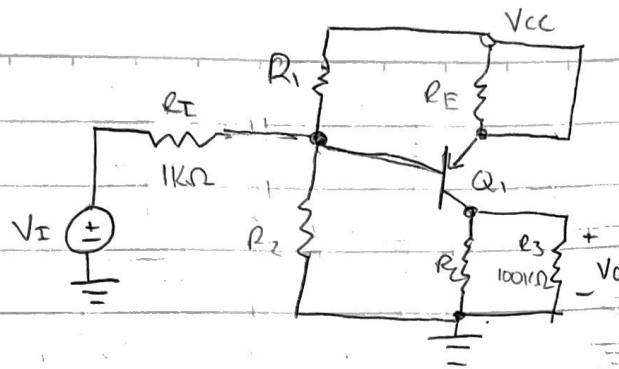
$$V_{EC} = 9V - 2.1071V - 6.864V$$

$$\therefore V_{EC} = 0.0651V$$

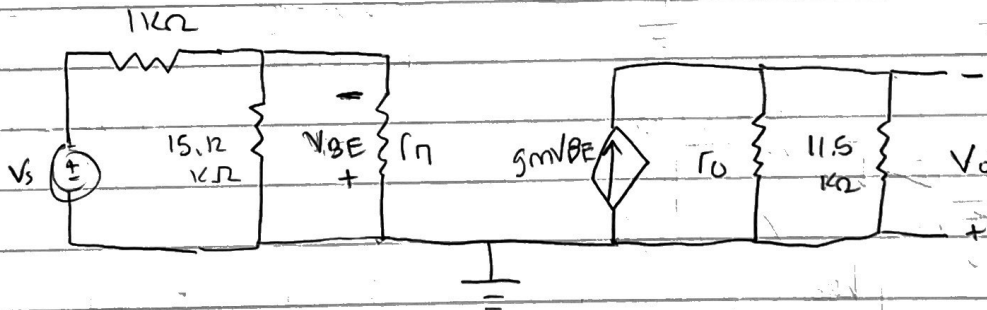
b) Q-point : (0.528mA, 0.651V)

2. AC Analysis →

a)



b)



c) $V_A = 100V$

d) $V_{EB} = \frac{15.12k\Omega \parallel 6.67k\Omega}{15.12k\Omega \parallel 6.67k\Omega + 1k\Omega} V_s$

$$g_m = \frac{q I_C}{kT} = \frac{0.528mA}{0.0259} = 0.0204 S$$

$$\therefore V_{EB} = \frac{4.4k\Omega}{5.6k\Omega} = 0.821 V_s$$

$$r_\pi = \frac{\beta_0}{g_m} = \frac{135}{0.0204} \approx 6.67k\Omega$$

$$-V_O + g_m V_{BE} \times R_L = 0$$

$$V_O = g_m V_{BE} \times R_L = 181.571 V_s$$

$$R_L = r_o \parallel 11.5k\Omega = 10.84k\Omega$$

$$r_o \approx \frac{V_A}{I_C} = \frac{100V}{0.528mA} \approx 189.39k\Omega$$

$$V_O/V_s = 181.571$$

3. Pierret Problem 16.4 (a) - (a)

$$a) \Phi_F (KT/q) = -\ln(N_D/N_i) = -11.51$$

$$\Phi_F (V) = -11.51 (KT/q) = -0.298 V$$

$$b) W = \left[\frac{2K_s \epsilon_0}{-q N_D} (2\Phi_F) \right]^{1/2} = 0.882 \mu m$$

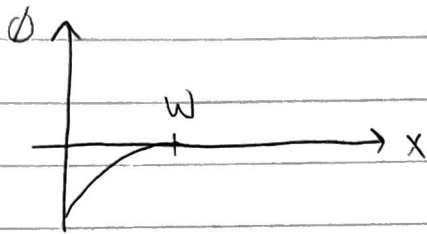
$$c) E_s = \frac{-q N_D}{K_s \epsilon_0} W = -1.35 \times 10^4 \text{ V/cm}$$

$$d) V_0 = 2\Phi_F + \frac{K_s x_0 E_s}{K_0} \quad ; \quad \Phi_s = 2\Phi_F$$

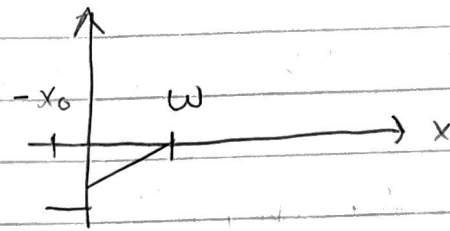
$\therefore V_0 = -1.0 V \rightarrow$ related to fig. 16.10 b/c the parameters are the same $\therefore$ the result should correspond to the onset of inversion.

4. Perret, 1b.7

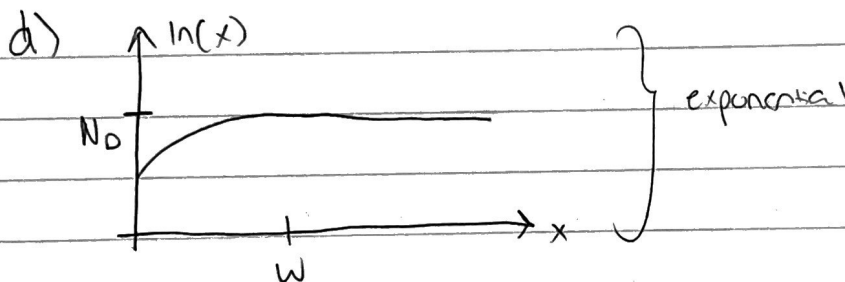
a) Sketch Φ



b) Sketch $\mathcal{E}$



c) Yes b/c inside the semiconductor, EF is not dependent on x



e) The electron concentration is about $n_i \approx 10^{10} / \text{cm}^3$

$$f) N_D \approx n_{\text{bulk}} = n_i e^{[E_F - E_i(\text{bulk})]/kT}$$

$$\hookrightarrow \therefore N_D = (10^{10}) e^{0.29/0.0259} = 7.29 \times 10^{14} / \text{cm}^3$$

$$g) \Phi_s = (1/q) [E_i(\text{bulk}) - E_i(\text{surface})] = -0.29 \text{ V}$$

$$h) V_0 = \Phi_s - \frac{K_s X_0}{K_0} \sqrt{\frac{2q N_D}{K_s \epsilon_0} (-\Phi_s)} \quad \therefore \text{plugging in values we get } V_0 \approx -0.78 \text{ V}$$

$$i) V_0 = \Delta \Phi_x + \Phi_s \rightarrow \Delta \Phi_x = V_0 - \Phi_s = -0.49 \text{ V}$$